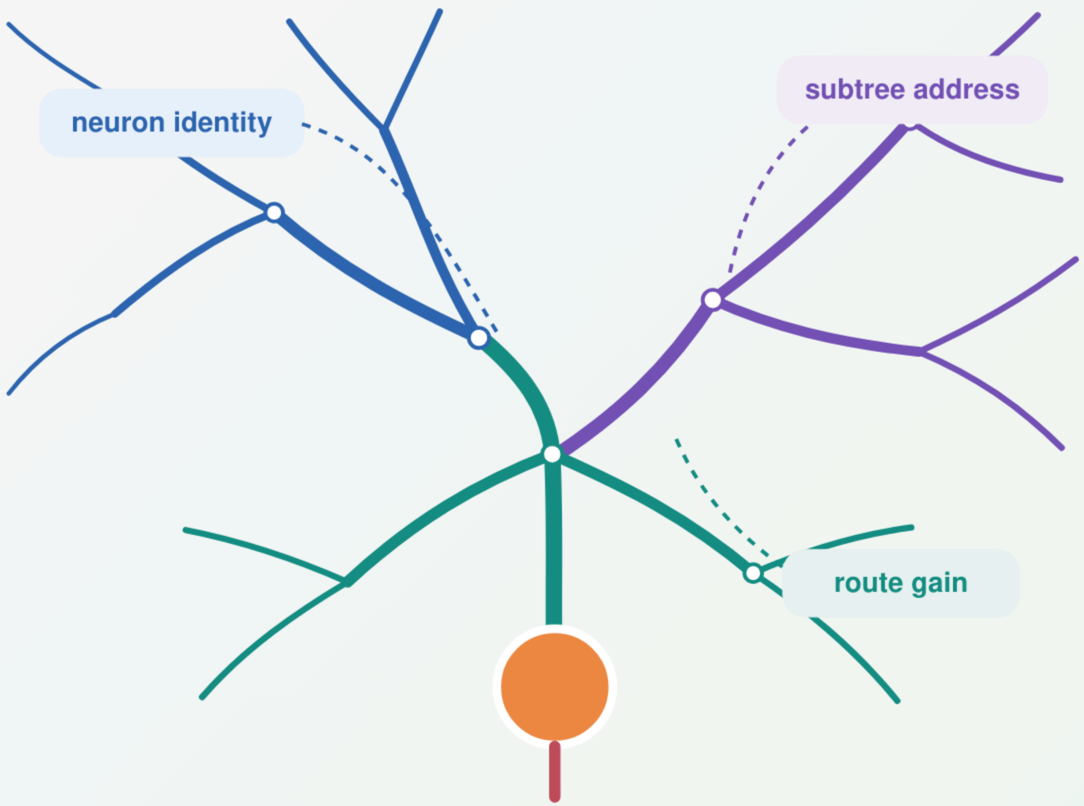
When can dendritic structure help local credit assignment?

From backpropagation to neuron coordinates, dendritic addresses, conductance-dependent route gain, and an alignment boundary



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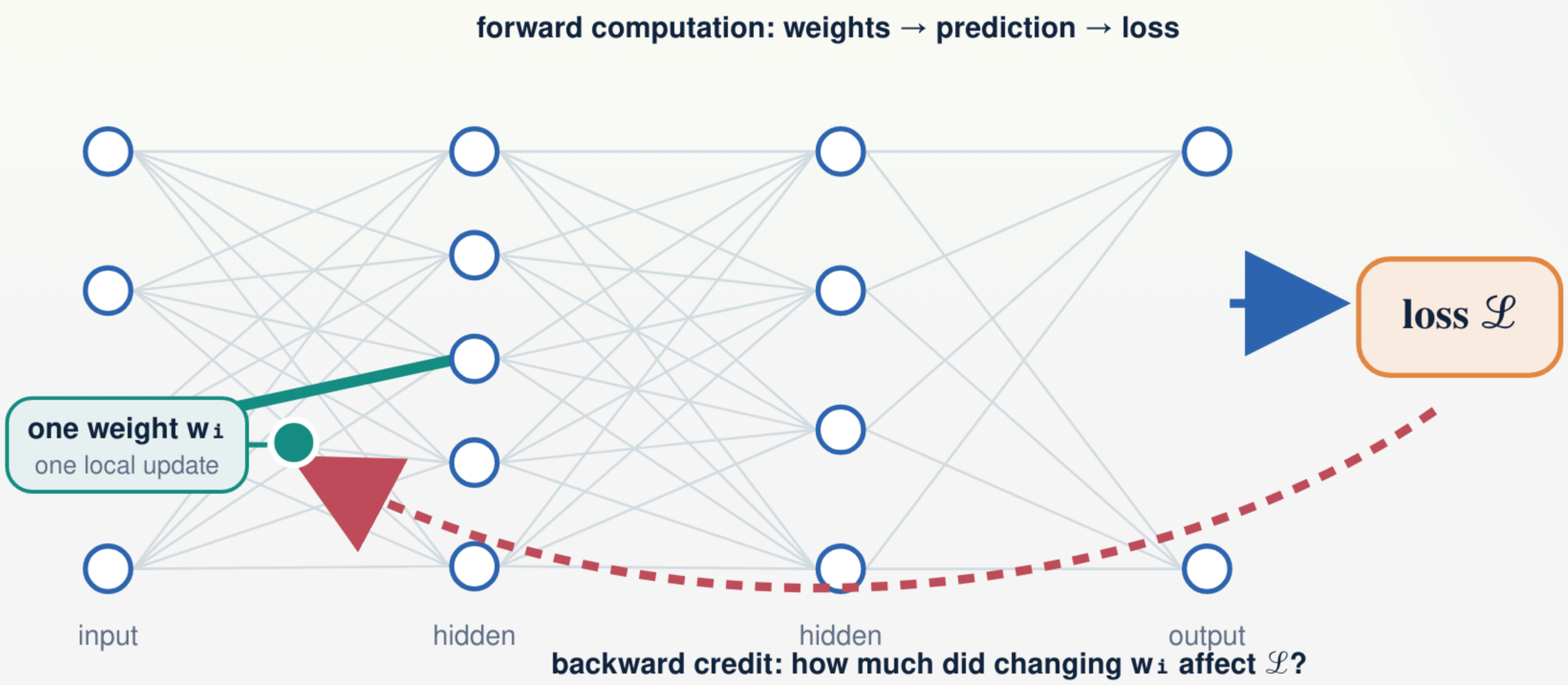
THE PROBLEM

Learning changes network weights; credit assigns each change



credit must specify

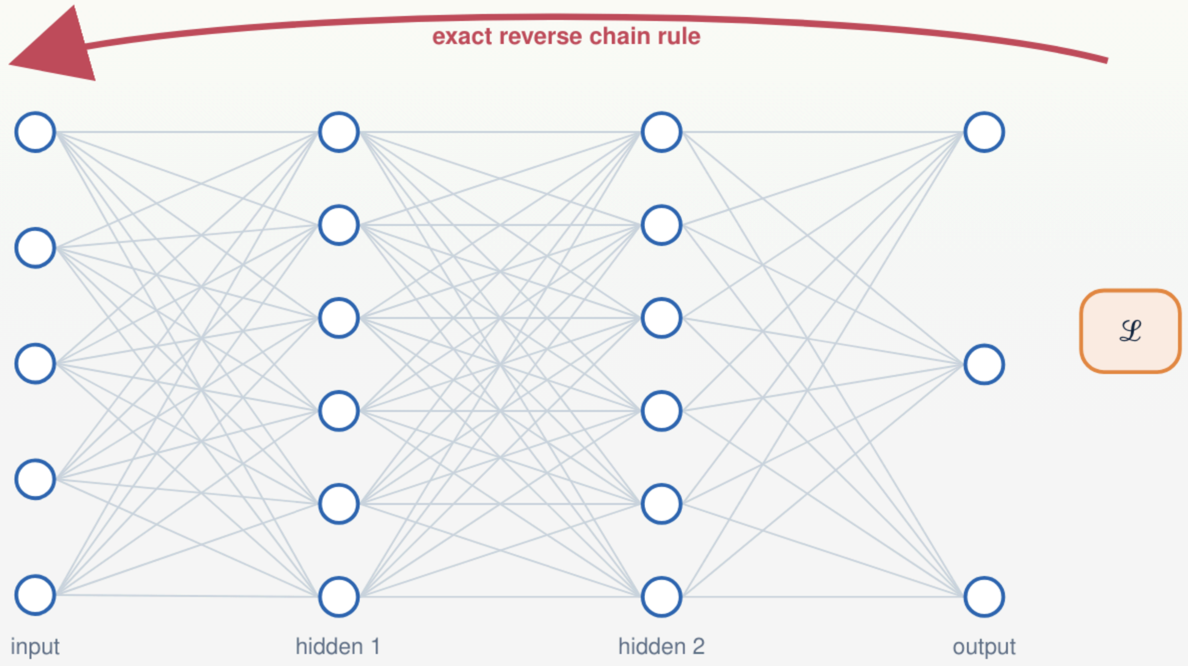
- which neuron?
- which location?
- sign and magnitude?

$$\mathscr{L} = \mathscr{L}(f(x;\, w),\, y) \quad \cdot \quad \Delta w_i = -\eta\, \partial\mathscr{L}/\partial w_i$$

→ **A global objective must be converted into a destination, sign, and magnitude for every trainable weight.**

THE MATHEMATICAL REFERENCE

Backpropagation defines the exact neuron-specific learning signal



$$z_u = \Sigma_{i \in in(u)} w_i x_i + b_u, \quad y_u = f_u(z_u)$$

$$\delta_u \equiv \partial\mathcal{L}/\partial y_u = \Sigma_{v:u \to v} w_{uv} f'_v(z_v)\delta_v$$

δ_u is exact task information assigned to neuron **u**—not yet to one weight.

fixed/random feedback	feedback alignment; direct feedback
inferred targets or states	equilibrium propagation; predictive coding
eligibility + learning signal	three-factor rules; e-prop
dendritic teaching signals	somato-dendritic and apical-error models

→ **Backpropagation is the information reference; biological theories differ in how the returned signal is produced.**

Rumelhart et al. (1986); Lillicrap et al. (2016); Scellier & Bengio (2017); Bellec et al. (2020)

Local learning preserves eligibility and approximates the returned signal

$$\partial\mathscr{L}/\partial w_i = \underbrace{x_i\, f'_u(z_u)}_{\text{local eligibility } e_i} \times \underbrace{\delta_u}_{\text{exact neuron signal}}$$

$$\Delta w_i = -\eta\, e_i\, \delta^{avail}_u$$

one layer-wide scalar m

Every neuron receives the same task-dependent coordinate.

one signal δ^{avail}_u per neuron

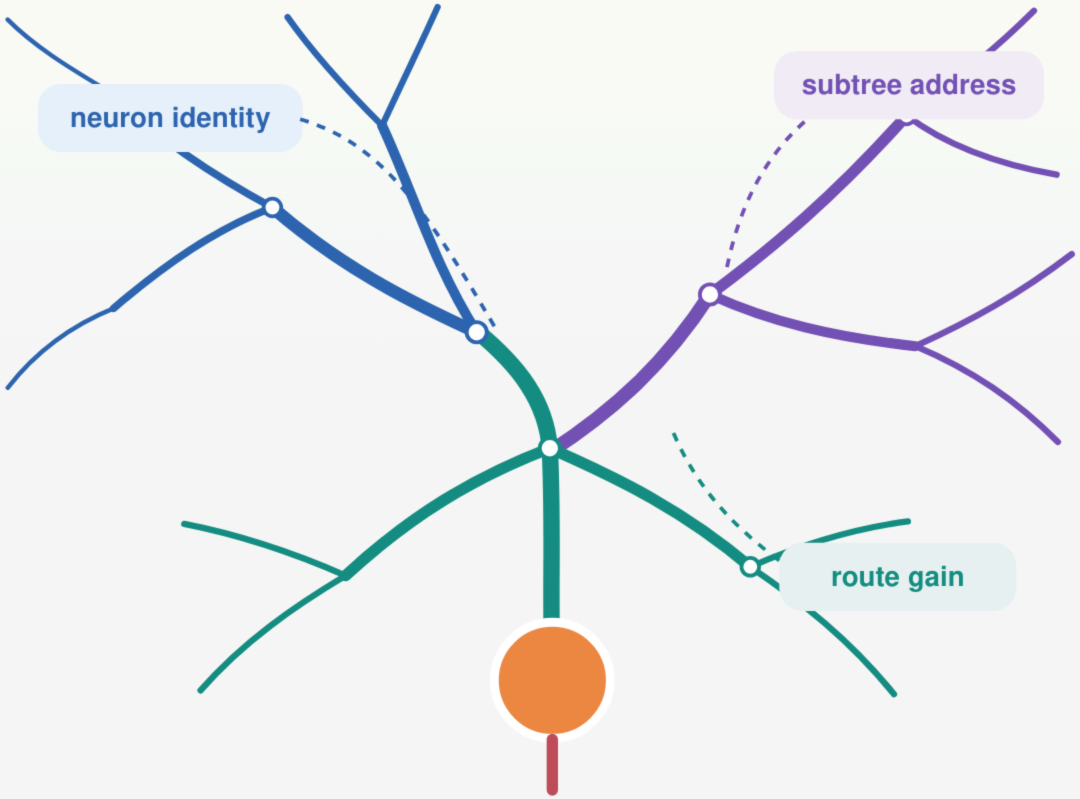
Feedback identifies the postsynaptic neuron while the eligibility remains weight-specific.

A shared scalar does not equalize the weights: each update still contains a different eligibility e_i.

→ **Locality determines where an update is computed; feedback bandwidth determines what task information it can use.**

Werfel, Xie & Seung (2005); Frémaux & Gerstner (2016)

Dendrites turn one neuronal signal into four testable routing questions



network weight w_i → synaptic conductance g_i on compartment n

$$\hat{\delta}V_u = A_u c_u, \quad A_u \in \mathbb{R}^{N_u \times K}, \quad c_u \in \mathbb{R}^K$$

A_u says where each feedback channel is delivered; c_u contains the K signed values available on the current example.

1 · coordinate + ownership
Which neuron—and which arbor—receives each signal?

2 · dendritic address
Which compartment or subtree receives it?

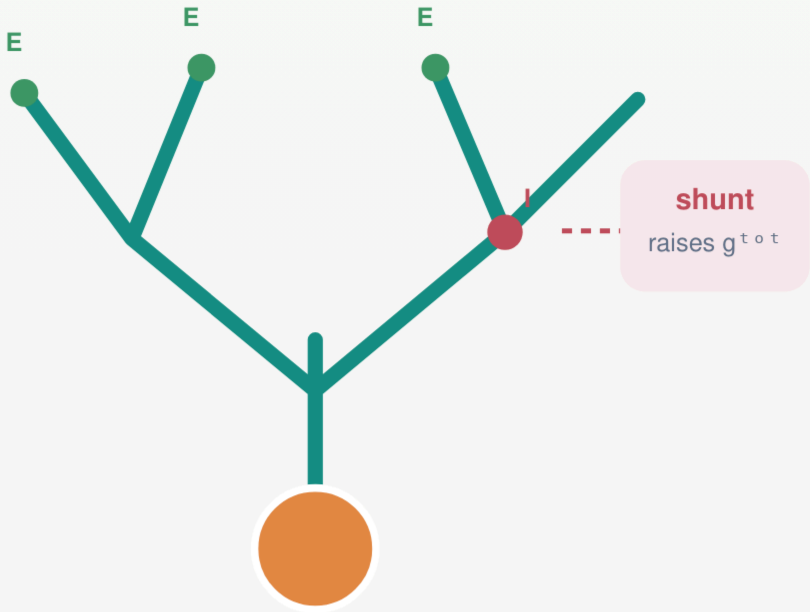
3 · route gain
How strongly does it reach that destination?

4 · task–route alignment
Does the available route span match the credit demanded by the task?

The experiments test coordinate, address, gain, and alignment separately instead of treating “dendrites” as one intervention.

DIRECTED-TREE STEADY STATE

Conductance makes dendritic voltage a normalized quotient



$G^E_n=\Sigma g^E_j x^E_j \ge 0$

$G^I_n=\Sigma g^I_j x^I_j \ge 0$

E and I are distinguished by reversal potentials E_E and E_I, not by negative conductance.

$$g^{tot}_n = g^L_n + G^E_n + G^I_n + G^{child}_n, \quad R^{tot}_n = 1/g^{tot}_n$$

$$V_n = R^{tot}_n(g^L_n E_L + G^E_n E_E + G^I_n E_I + J^{child}_n)$$

additive current

Changes the numerator without changing total conductance.

shunting conductance

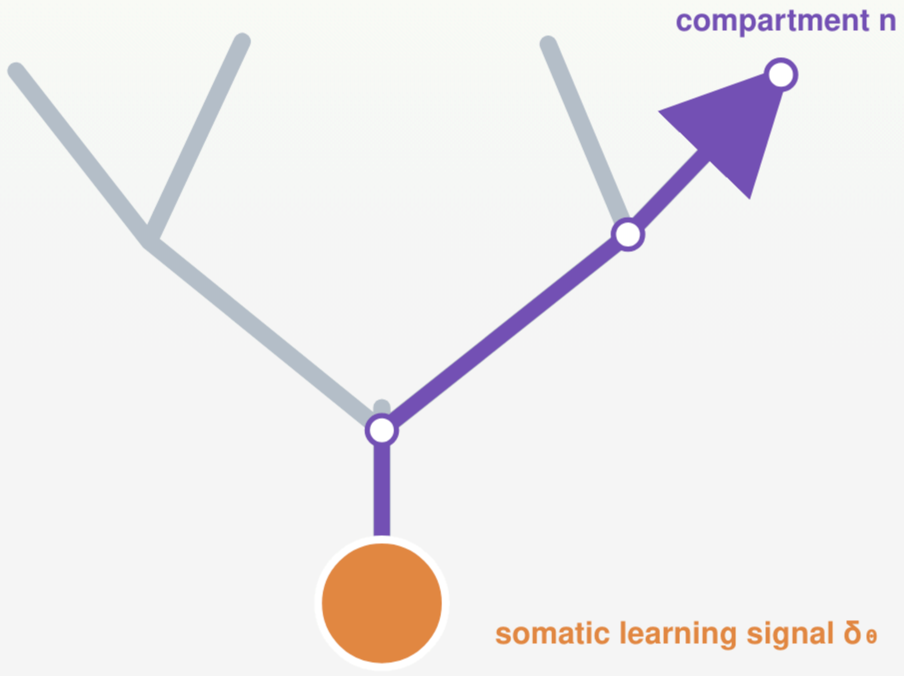
Raises g^{tot}_n and lowers R^{tot}_n, even when net shunt current is small.

$V_n \approx E_I \Rightarrow g_I x_I(E_I - V_n) \approx 0$, but $g_I x_I$ still changes the denominator.

→ **Shunting changes local sensitivity because conductance appears in the voltage denominator.**

Holt & Koch (1997); Chance et al. (2002); Gidon & Segev (2012)

The exact dendritic gradient is local eligibility × transported compartment error



one somatic error $\delta^V_{0,u}$
is transported into a field over compartments

$\delta^V_{n,u} \equiv \partial\mathcal{L}/\partial V_n$: exact error assigned to compartment n of neuron u

$$\partial\mathcal{L}/\partial g_i = \underbrace{x_i R^{tot}_n(E^{rev}_i - V_n)}_{\text{local dendritic eligibility } e^{den}_i} \times \underbrace{\delta^V_{n,u}}_{\text{transported compartment error}}$$

$$\delta^V_{n,u} = \delta^V_{0,u}\tilde{\alpha}_n$$

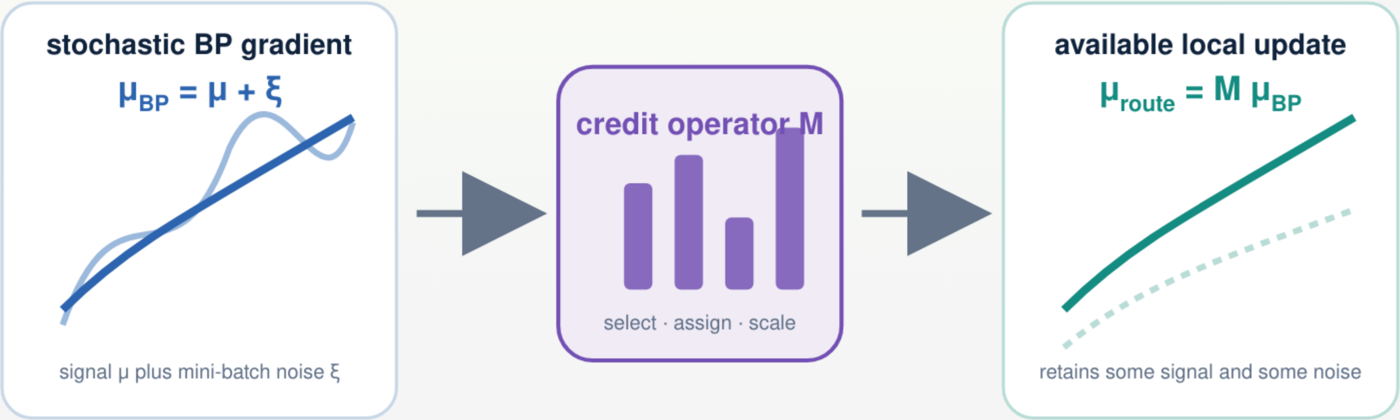
$$\tilde{\alpha}_n = \prod_{(i\to k) \in \mathrm{path}(n\to 0)} f'_i(V_i)\, R^{tot}_k\, g^{den}_{i\to k}$$

Directed tree: one exact path product. **Reciprocal cable:** the same field is obtained from the steady-state adjoint.

→ **Feedback must deliver a compartment-specific field; the synaptic eligibility itself is local and exact.**

Almeida (1987); Pineda (1987); Schiess, Urbanczik & Senn (2016)

A feedback pathway selects, assigns, and scales stochastic credit



$\mu_{BP} = \mu + \xi, \quad E[\xi] = 0, \quad \mathrm{Cov}(\xi) = \Sigma$

$\mu_{route} = M\mu_{BP}, \quad w_+ = w - \eta\mu_{route}$

M = I: unrestricted backpropagation of the stochastic gradient. Restricted routes change the span, assignment, or gain of the available update.

The operator M lets one theory describe scalar, neuron-specific, subtree-targeted, and exact compartment feedback.

Operator utility balances retained signal, update cost, and noise

OPERATOR UTILITY · POSITIVE TASK ALIGNMENT REQUIRED

$$U(M)=\frac{[\mu^{T}M\mu]^2}{2L_{sm}[\|M\mu\|^2+\mathrm{tr}(M\Sigma M^{T})]}$$

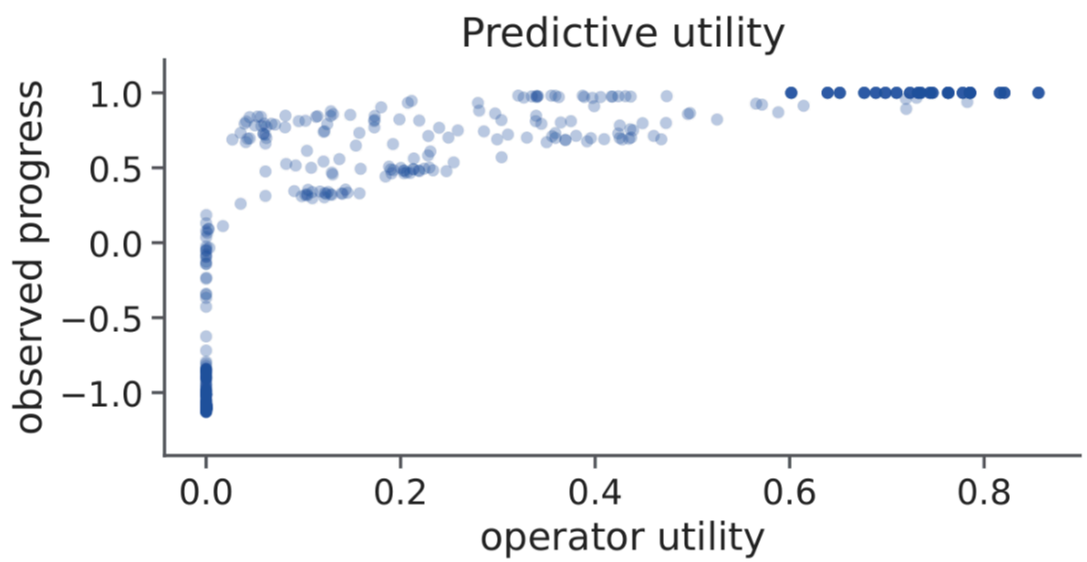
task-aligned signal · finite-step update cost · admitted stochastic noise

$\rho_S = 0.937$

utility versus observed norm-matched one-step progress across 540 trained conditions

$\rho_S = 0.916$

utility versus final trained accuracy

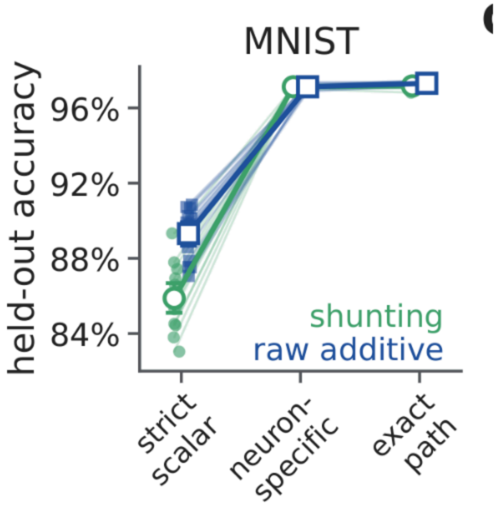


Exact for an isotropic quadratic with Hessian $L_{sm}I$; otherwise a curvature bound and one-step smoothness guarantee. It predicts large contrasts and within-family progress—not every trajectory-accrued difference.

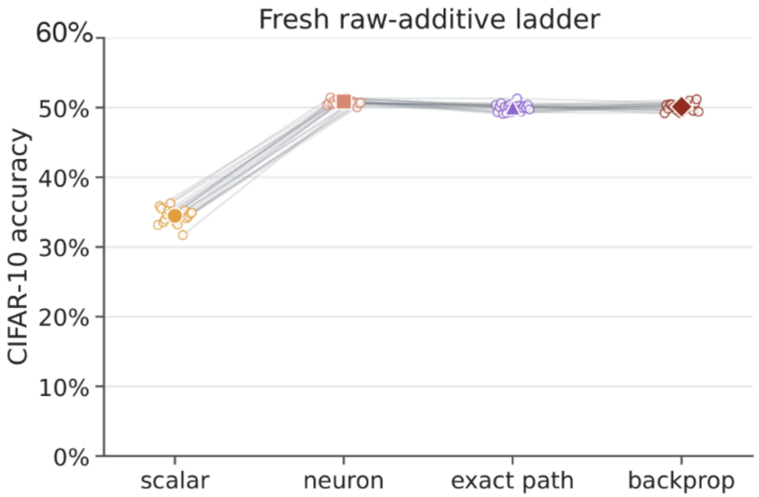
→ **Restricted routing helps only when its retained, aligned signal compensates for the signal it discards and the noise it admits.**

Neuron-specific feedback dominates standard image tasks

MNIST · green = shunting · blue = additive



Flattened CIFAR-10 · additive tree · “exact path” = exact compartment field



+7.8 to +16.4 pp
strict scalar → one distinct feedback coordinate per neuron

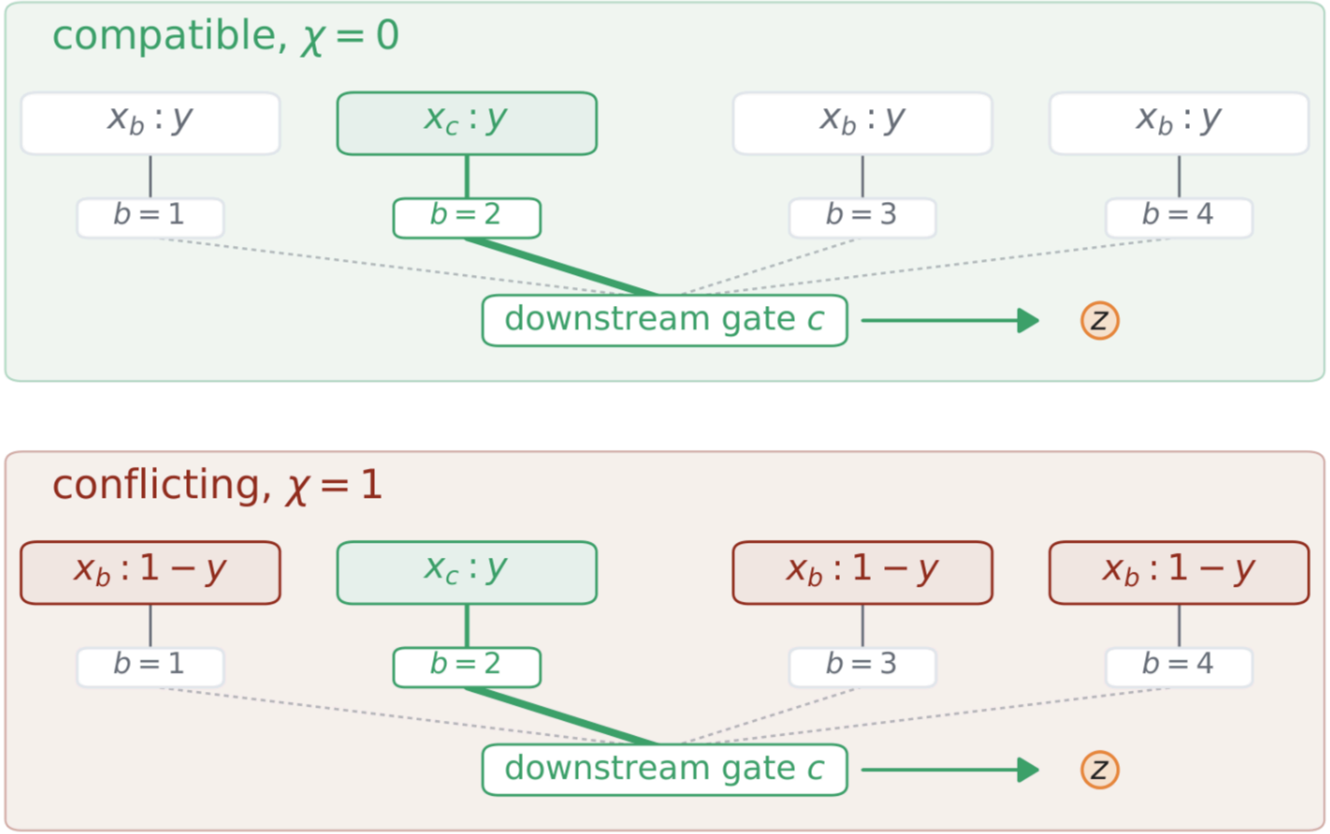
−0.86 to +0.19 pp
neuron-specific coordinate → exact compartment field

≈ backprop
exact field on flattened CIFAR-10, within the predefined ±1-point margin

Most of the gain comes from preserving distinct feedback coordinates across neurons; exact within-tree resolution adds little.

Paired-seed means; CIFAR exact field and BP are equivalent within the predefined ±1-point margin.

Credit conflict creates a demand for branch-specific signals



input

All B branches receive a Fashion-MNIST image and form nonzero eligibility.

forward selector

Context c selects the branch whose image determines the target.

conflict dose χ

Nonselected images vary from class-compatible (χ=0) to opposite-class (χ=1).

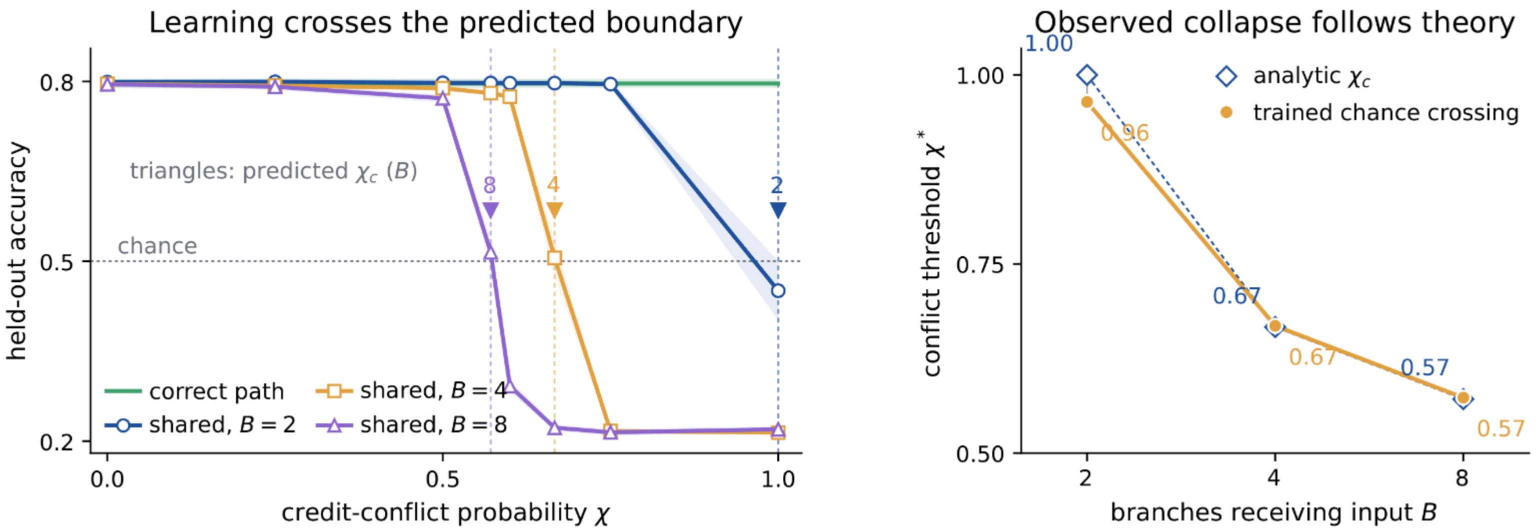
$$d^{\text{branch}}{}_b = N^{-1}\Sigma_t\delta_t\mathbf{1}[c_t{=}b]x_{t,b}$$

$$d^{\text{shared}}{}_b = N^{-1}\Sigma_t\delta_t B^{-1}x_{t,b}$$

N: trials · **δ_t**: downstream logit gradient · **1[·]**: indicator
d_b: mean update direction for branch b

At zero conflict, one shared signal is sufficient; at high conflict, different branches require opposite updates.

Shared credit fails at the predicted conflict boundary



$\lambda_{shared}=B-2\chi(B-1) \rightarrow \chi_c=B/[2(B-1)]$

35–58 pp

branch-specific gain over shared feedback at full conflict

causal control

Cyclically deranged routes fail despite identical rank and sparsity.

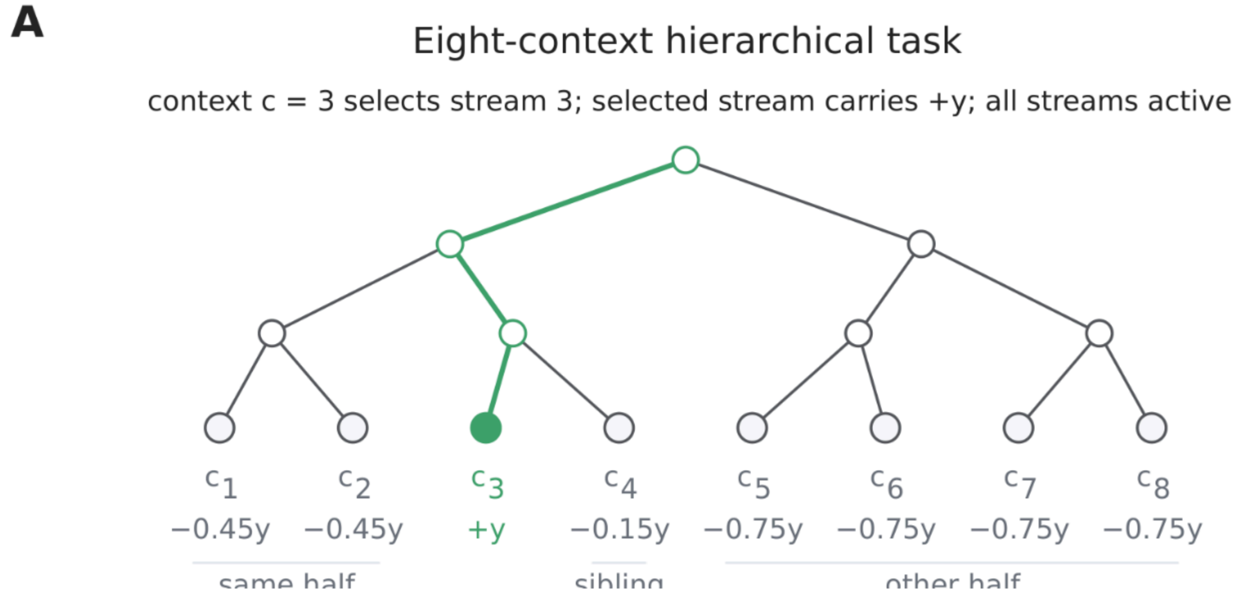
implementation control

Analytic BP, the correct route, and a gated-point calculation are equivalent forms of the same routing matrix.

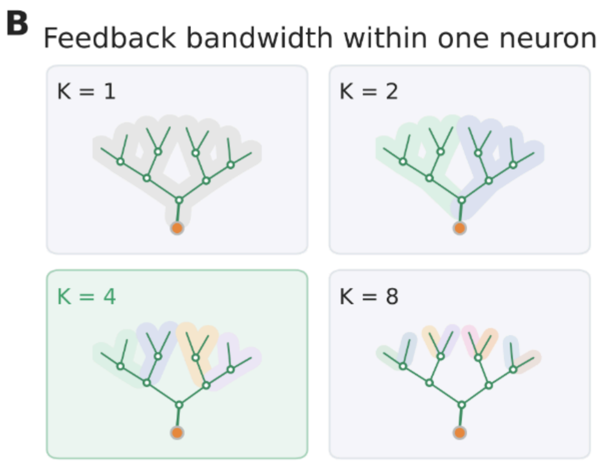
→ **The task establishes a need for an address—not a unique need for dendritic material.**

B = 2, 4, 8 branches; all empirical collapse points track the analytic boundary.

Nested tasks ask whether a few subtree addresses are efficient



$$\delta^{V,\,\mathrm{avail}} = A_K\beta, \quad \mathrm{rank}(A_K)=K$$

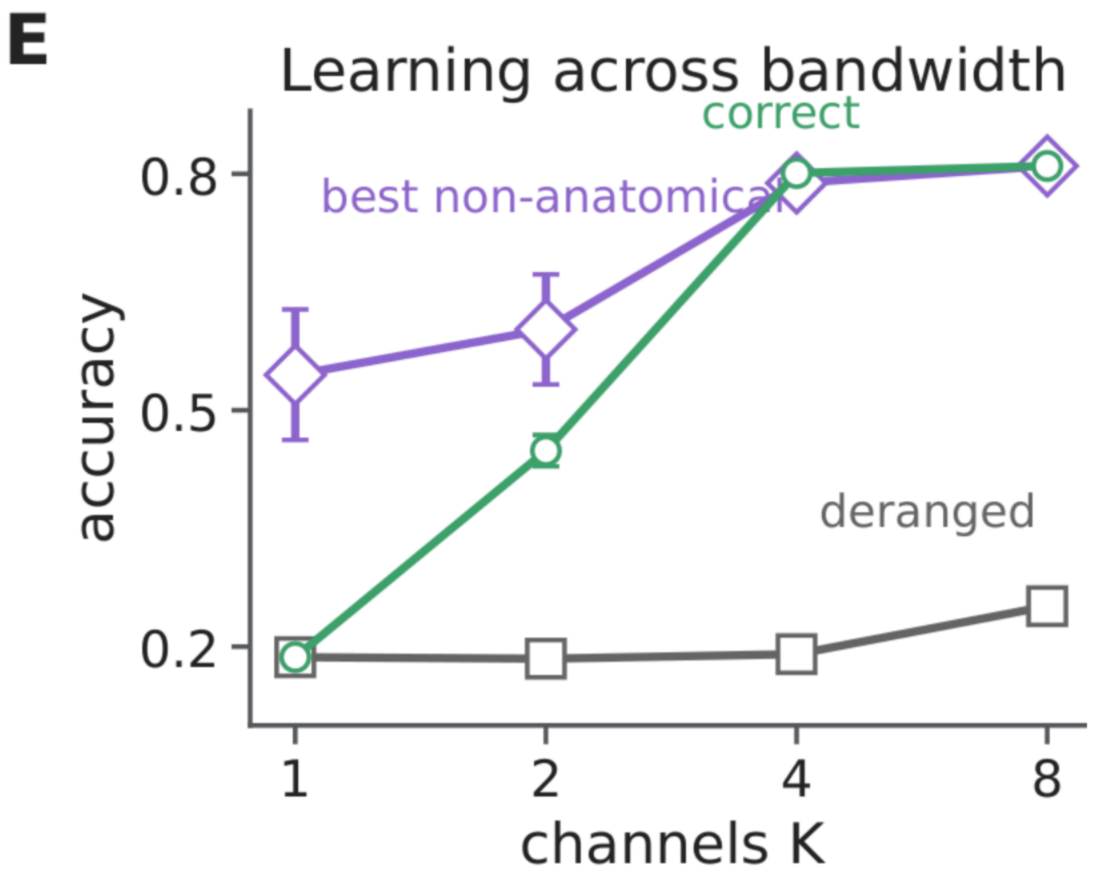


matched controls

Rank, sparsity, parameter count, and forward resources are held fixed; only route assignment, basis, or topology changes.

→ **Intermediate K tests whether tree-structured feedback is a useful low-dimensional basis for task credit.**

Subtree addresses help—but fine topology adds only a narrow gain



+61.1 pp

correct ancestry assignment over cyclic derangement at the same K=4 bandwidth

+1.27 pp

correct ancestry over the strongest matched non-anatomical low-rank control at K=4

against the matched low-rank basis

Ancestry loses at K=1,2; wins only at K=4; and ties at full rank K=8. Rewiring removes the intermediate advantage.

equivalence

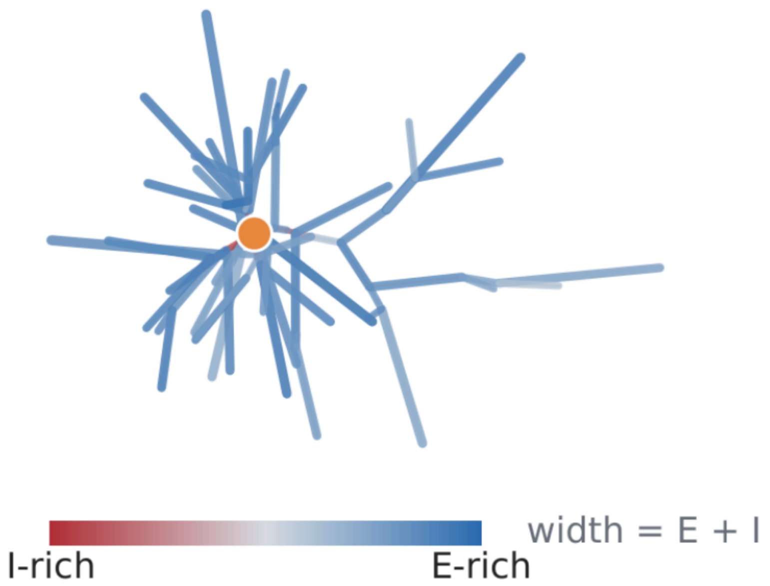
Dendritic, grouped-point, and gated-point implementations coincide for the same routed field.

→ **The large effect is the value of within-neuron addresses; the anatomy-specific effect is conditional and modest.**

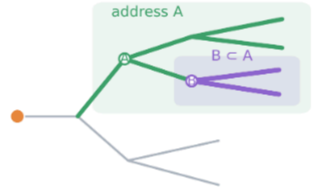
Twenty paired seeds; K=4 matched-topology contrast survives BH correction.

Real arbors provide sparse candidate routes, mostly through coarse geometry

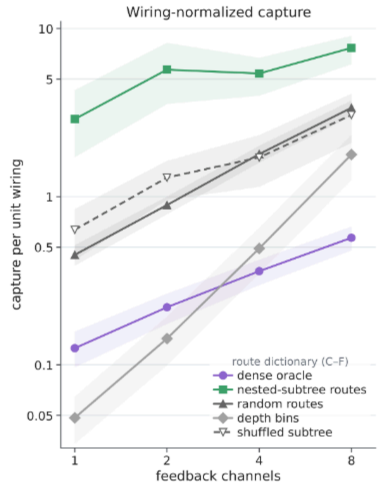
measured MICrONS morphology with mapped E/I contacts



branch points define nested route supports



$C_A(q)=\|P_A q\|^2/\|q\|^2$ fraction of field energy in the route span



85%
of dense field capture

≈ 7%
of dense route-matrix connections

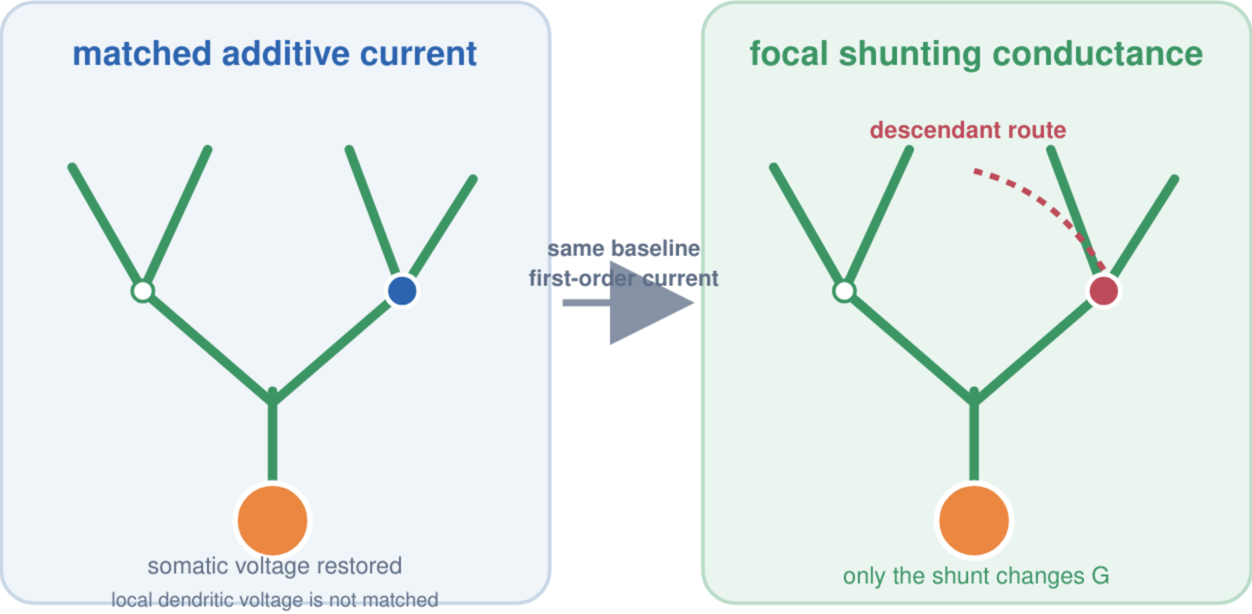
14.2×
dense capture per connection

≈ 2.7×
over a density-matched shuffled dictionary

Route-matrix connections are not cable length or energy. The ordering replicated in 47 held-out cells, and the subtree advantage was positive in 10/10 quality-controlled cells from a second MICrONS mouse. These are modeled fields, not observed task gradients.

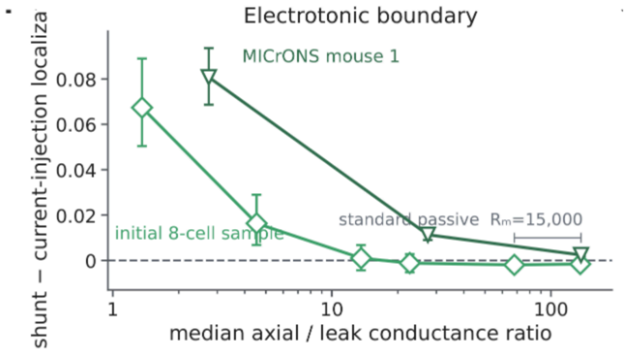
Measured morphology supplies a sparse route dictionary; most capacity comes from branch depth and coarse topology.

Focal shunting changes descendant credit only in permissive regimes



$$\partial \log \alpha^{\mathrm{cond}}_n / \partial G^{I}_k = -R^{\mathrm{tot}}_k \, \mathbf{1}[k \in (n)]$$

A shunt directly attenuates only routes descending through compartment k; the effect scales with local input resistance.



standard passive calibration

At R_m=15,000 Ω cm², the shunt-minus-current localization contrast is effectively zero.

high-conductance regime

Descendant-localized changes emerge and survive active-channel extensions.

Shunting is a state-dependent route-gain mechanism, not a generic learning advantage.

Baseline first-order focal current is matched and somatic voltage restored; local dendritic voltage is not matched.

Measured visual responses show no morphology-specific alignment

measured presynaptic responses

→

mapped conductances on each reconstructed target arbor

→

held-out postsynaptic-response prediction

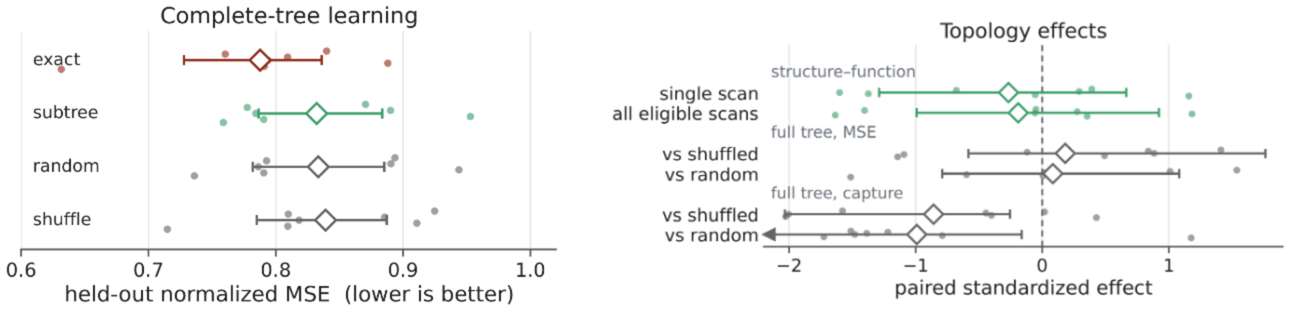
question

Does the measured input–output relation align more strongly with nested subtrees than with matched alternative route dictionaries?

inferential scope

Seven target cells from one MICrONS mouse; exact compartment errors are the reference.

Held-out learning and topology-minus-control effects



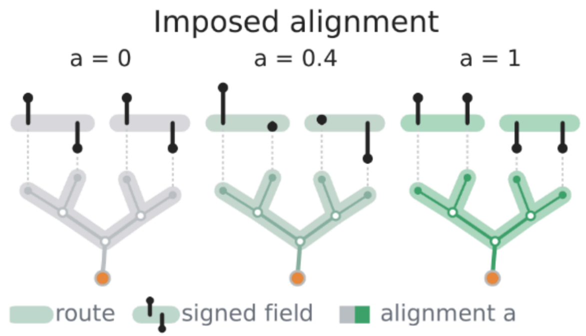
Nested subtrees do not outperform random or site-shuffled routes for held-out prediction, field capture, or within-arbor structure–function similarity.

Measured anatomy supplies candidate routes, but these visual responses provide no evidence that the task uses them preferentially.

The same anatomical routes capture task credit aligned to their span

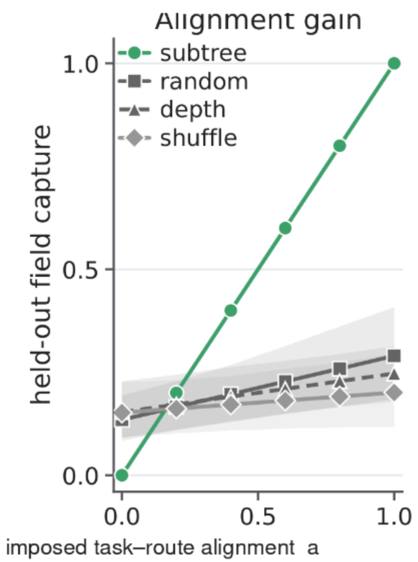
$$\varphi(a) = \sqrt{a}\, u_{\parallel} + \sqrt{(1-a)}\, u_{\perp}, \quad a \in [0,1]$$

$u_{\parallel} \in \mathrm{col}(A)$, $u_{\perp} \perp \mathrm{col}(A)$, $\|u_{\parallel}\| = \|u_{\perp}\| = 1$



controlled quantity

Anatomy, field energy, curvature, and route count are fixed; only alignment with the subtree span changes.



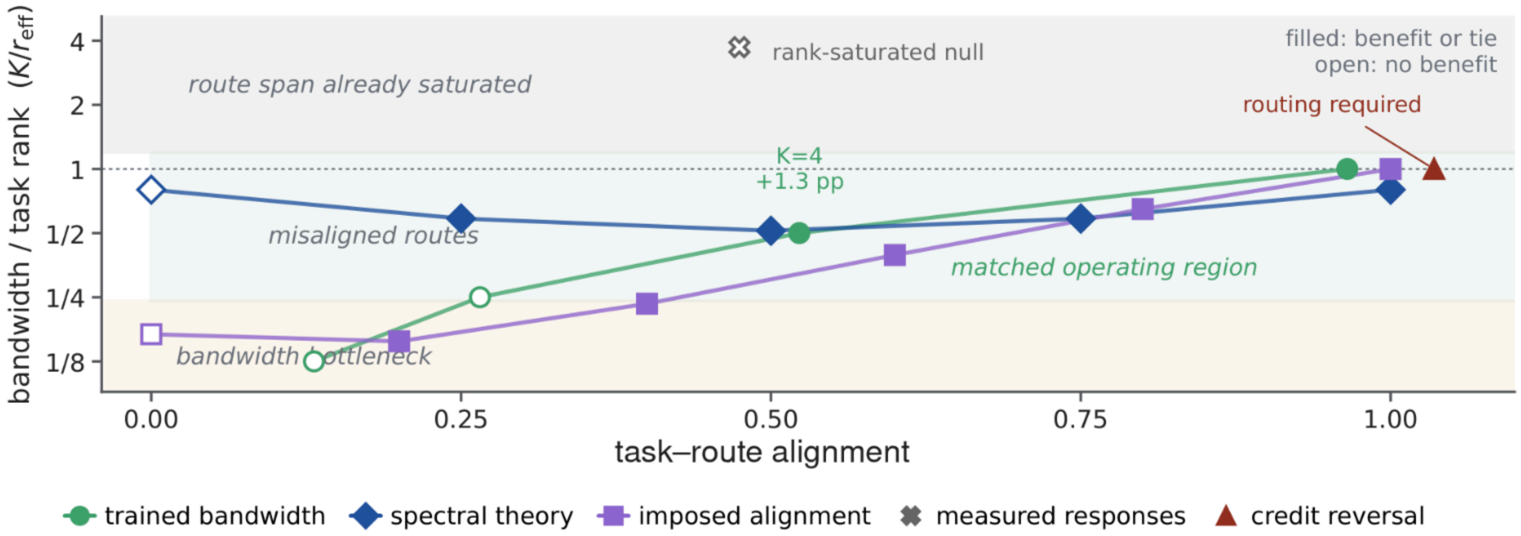
$$C_A[\varphi(a)] = a \quad \text{by construction}$$

n = 8 reconstructed cells

Controls test whether the gain is specific to the true subtree span. The manipulation proves conditional representational sufficiency—not trained learning or endogenous biological use.

Alignment is sufficient for representational capture in the model; endogenous morphology-specific alignment remains unestablished.

One alignment–bandwidth plane organizes the wins and nulls



low relative bandwidth

Too few independent coordinates: neuron identity or branch address is the bottleneck.

aligned intermediate bandwidth

Restricted routes can retain task credit while rejecting irrelevant dimensions.

full rank or weak alignment

Route capacity saturates, or available structure does not match the task.

vertical: relative bandwidth K/r_{eff}
horizontal: task–route alignment

Coordinates are estimated separately within each experiment. Regime tint and the $K/r_{\mathrm{eff}}=1$ boundary are theoretical, not fitted.

Useful route resolution requires both sufficient feedback bandwidth and alignment between the task-credit field and the available routes.

TAKE-HOME

Coordinate → address → gain, all conditional on task–route alignment

coordinate
which neuron?

→

address
which subtree?

→

gain
how strongly?

enabled by **alignment + bandwidth**

1 · neuron-specific coordinates come first

Across MNIST and flattened CIFAR-10, preserving distinct feedback across neurons closes most of the scalar-to-exact gap.

2 · addresses matter when local updates conflict

Branch-specific signals become necessary when simultaneously active subtrees require different or opposite changes.

3 · topology helps only in a matched regime

Nested routes add a modest advantage at intermediate aligned bandwidth; reconstructed arbors supply sparse candidate routes.

4 · conductance regulates gain conditionally

Focal shunting localizes modeled credit only in permissive states, and endogenous morphology-specific use remains unestablished.

Dendrites are a conditional substrate for routing local credit—not a general replacement for backpropagation.

→ **The central contribution is a predictive boundary map: when restricted dendritic routes help, when they do not, and why.**